Reg. No. | | | | | | | | | | | [14EE36]

THIRD SEMESTER B.TECH EXAMINATION DECEMBER-2015

ELECTRICAL MEASUREMENTS AND MEASURING INSTRUMENTS

Time:3 Hours **Maximum Marks:70**

Instructions: Answer five full questions, choosing at least TWO from each part.

PART-A

1. a. Find the relationship between electrostatic and electromagnetic units. **06**

   b. The resonant frequency of an AC series circuit by **08**

   $f_r = \frac{1}{2\pi} L^a C^b$ (L is the inductance), find the values of a and b.

2. a. With the help of neat diagram, explain the operation of megger **08**

   b. A sheet of bakelite 4.5mm thick is tested at 50Hz between electrodes 0.12m in diameter. The Shearing bridge employs a standard air-capacitor C_2 of 106 pF, a standard non inductive resistance R_4 of $100/\pi$ Ω in parallel with a variable capacitor C_4 and a non inductive variable resistance R_3. Balance is obtained with C_4=0.5 μF and R_3=260 Ω. Calculate the capacitance, power factor and relative permittivity of the sheet. **06**

3. a. What are shunts and multipliers? Derive an expression for both with reference to meters used in electrical circuits. **06**

   b. Draw the equivalent circuit and phasor diagram of a current transformer. Also derive the expressions for ratio and phase angle errors. **08**

4. a. With the help of a neat diagram, explain the construction and working of a single phase induction type energy meter. **08**

   b. A wattmeter is connected to measure the power consumed by a load, which draws a current of 20A and 250V. The power factor of the load is 0.8 lagging the current coil having impedance of (0.08+j0.4)Ω. The resistance of the potential coil is 6000Ω. Find the error in the reading of the wattmeter for its two possible connections to measure the power. **06**

PART-B

5. a. Explain with the help of block diagram, the working of true RMS reading voltmeter. **07**

   b. Write a note on phase sequence indicator. **07**

6. a. Explain the working of digital storage oscilloscope. Also explain the different modes of operation. **10**

   b. Explain the measurement of frequency using Lissajous patterns. **04**

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7.	**a.**	Prove that the gauge factor of strain gauge is given by $K = 1 + 2\mu$, where μ is the Poisson's ratio.	**06**
	b.	With a neat sketch explain the construction and working of linear variable differential transformer transducers.	**08**
8.	**a.**	Differentiate between pulse and square wave generator.	**04**
	b.	Write short notes on the following;	**10**
		i. LED	
		ii. LCD	